Prediction-driven reinforcement learning for sensing system scheduling under a power constraint

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Abstract

Multi-sensor systems cannot always operate all measurement channels simultaneously under a prescribed power budget. We propose PD-PPO, a prediction-driven reinforcement-learning framework that schedules executable channel subsets to improve downstream multistep forecasts. A masked categorical PPO policy chooses among subsets satisfying power, startup, and minimum activation rules. It uses recent partial observations, channel ready-sampling age, runtime, previous selections, and early event-warning scores; a forecaster fitted on an earlier partition supplies the training loss. We evaluate PD-PPO in a six-channel cold-region monitoring simulation where particle, flux, and thermal regimes favor different specialists. Across 24 final seeds, including an unchanged 22-seed post-pilot replication, PD-PPO reduces mean forecast loss by 10.1% and an equal-weight regime score by 7.9% against validation-selected fixed schedules, winning both comparisons in every seed. Continuous replay over all 5,242 scoreable final-partition epochs preserves 24/24 wins. PD-PPO also outperforms AoI-priority, round-robin, random, and one-step forecast-greedy schedules in every seed, and

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matched Double-DQN on the macro score in every seed. Its four-specialist actions pass non-fixed and non-cyclic checks in all seeds and produce no warm-up aborts. An independent ridge forecaster retains a positive macro comparison against its own validation-selected fixed schedule in 23/24 seeds. Handcrafted warning-score and exact-label references remain competitive, showing that event information captures much of the available scheduling value. These results support forecast-aligned masked reinforcement learning when a power budget forces different measurement choices across operating regimes.
Keywords: Sensing system scheduling, Power budget, Time series forecasting, Reinforcement learning, Adaptive sensing, Antarctic monitoring

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Abbreviations

AoI, age of information; AWS, automatic weather station; DRL, deep reinforcement learning; PD-PPO, prediction-driven proximal policy optimization; PPO, proximal policy optimization; SOC, state of charge.

1. Introduction

Multi-sensor systems cannot always operate all measurement channels simultaneously when deployed under a prescribed power budget. A scheduling policy must then choose which channels to activate over time while respecting startup requirements and minimum activation times. Cycling, age-of-information (AoI), and uncertainty rules provide practical choices, but they evaluate the availability or estimated quality of current measurements. They do not directly measure whether a selected observation improves a later forecast.

This distinction matters when the value of a channel changes with operating conditions. A particle sensor may be most useful around a transport event, a flux sensor during high mass transport, and a surface-temperature sensor during a thermal transition. The effect of an activation can also be delayed by sensor startup, the observation history used by the forecasting model, and the minimum time for which a selected channel remains active. The resulting problem is sequential: the best executable choice at the current epoch need not be the schedule that gives the lowest forecast loss over future epochs.

This paper proposes prediction-driven proximal policy optimization (PD-PPO), a reinforcement-learning framework for sensing system scheduling under a power budget. PD-PPO evaluates channel selections by downstream multistep forecast loss and represents each action as an executable channel subset. A feasibility mask removes choices that violate the current power, startup, or minimum activation rule before the PPO actor samples an action. The method therefore learns over the action set that the sensing system can execute, rather than learning unconstrained choices that are repaired after selection.

At each epoch, the policy receives recent sample-and-hold values and observation masks, channel ready-sampling age, sensor runtime state, the previous selection, budget and time features, and supplied event-warning scores. It does not receive the exact simulator event label during final execution. A fixed forecasting model supplies the training loss, so the policy is optimized for the stated downstream task without changing the forecasting model during reinforcement learning. Training uses labelled auxiliary signals from the policy-training partition to improve representation learning; the final action is always produced by the masked PPO actor.

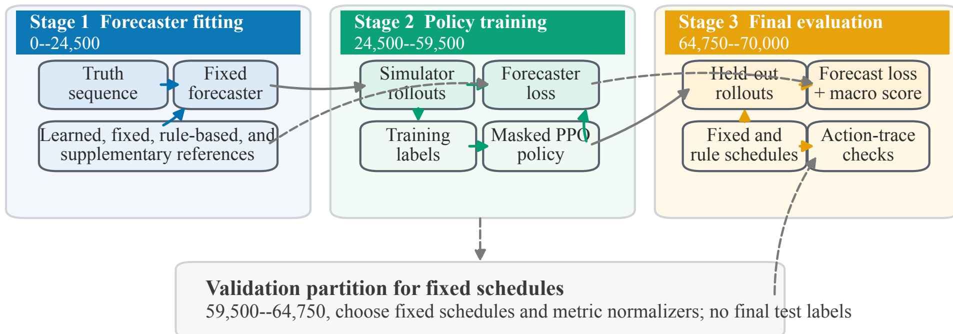

Figure 1. Prediction-driven sensing system scheduling. A forecasting model is fitted first and then held fixed while PD-PPO learns from the forecast loss produced by partial observation histories. Online feasibility masking admits only channel subsets that satisfy the current operating rules. Forecaster fitting, policy learning, validation, and final evaluation use separate chronological partitions.

The framework is evaluated in a controlled cold-region monitoring simulation motivated by Antarctic blowing-snow sensing. The system contains six logical measurement channels: one weather backbone that remains active and five specialist channels that compete for the remaining power. The budget admits one specialist at a time. Particle, flux, and thermal regimes favor different measurements, so a validation-selected fixed schedule provides a strong test of whether adaptive allocation is useful.

Across 24 evaluation seeds, including an unchanged 22-seed post-pilot replication, PD-PPO has lower mean forecast loss and a lower equal-weight regime macro score than the validation-selected fixed schedule in every seed. Mean forecast loss and the macro score fall by 10.1% and 7.9%, respectively. Replaying the frozen policies continuously over all 5,242 scoreable epochs in each final partition preserves 24/24 wins against the selected fixed schedule. PD-PPO also exceeds the AoI-priority, round-robin, random, and one-step forecast-greedy schedules in 24/24 comparisons. It exceeds matched Double-DQN on the macro score in 24/24 seeds and on mean loss in 23/24. A second frozen forecasting family preserves the positive macro comparison against its own validation-selected fixed schedule in 23/24 seeds. The supplied-warning rule and exact-label diagnostic show no detected mean difference from PD-PPO, indicating that explicit event information captures much of the available scheduling value.

The paper makes three contributions.

1. It formulates prediction-driven sensing system scheduling as a constrained sequential decision problem and establishes conditions under which forecast loss can favor a different schedule from AoI or uncertainty objectives.

2. It develops PD-PPO, which combines a multistep forecast-loss reward with categorical PPO over executable channel subsets and online feasibility masking for power, startup, and minimum activation rules.

3. It provides matched reward and learner controls, fixed and adaptive references, action-trace analysis, and independent-forecaster rescoring to distinguish adaptive scheduling value from a favorable fixed subset, a simple cycle, selected evaluation windows, or a particular forecasting model.

2. Background and Related Work

2.1. Resource-constrained sensing

Sensor scheduling under resource constraints has a long history in estimation, communication, and energy-aware monitoring. Representative objectives include covariance reduction, information gain, censoring utility, and age of information (AoI) (Shi et al., 2011; Kaul et al., 2012; Fernández-Bes et al., 2015; Qu et al., 2022; Alali et al., 2024; Ahdab et al., 2025; Jonah et al., 2026; Tran et al., 2026). These objectives are natural when the scheduler is judged by the freshness or uncertainty of the current state. The forecasting setting studied here keeps the same resource-constrained structure but changes the scoring target. A channel is valuable when it improves future predictions over the evaluation horizon.

That change matters for heterogeneous sensing systems. Background channels can be cheap and broadly informative, while specialist channels can be valuable only near particular physical regimes. A strong fixed subset is therefore an demanding reference. The evaluation must distinguish state-dependent scheduling from a fixed design, periodic rotation, and a privileged event-label reference.

2.2. Adaptive sensing and learning-based scheduling

The problem also connects to active perception and partially observable decision making, where observations are actions selected for a future task (Bajcsy et al., 2018; Lauri et al., 2023; Golovin and Krause, 2011). These lines of work motivate sensing decisions that change with state, but they do not specify how forecast-aligned specialist choices should be executed under explicit power, startup, and activation rules.

Table 1. Position of this work relative to nearby scheduling and sensing literatures.

Literature line	Typical optimization target	Scheduling structure	Difference addressed here
State-estimation and AoI scheduling	Covariance, estimation error, information age	Sensor selection under communication or energy limits	Multi-step forecast loss is the scoring target.
Active perception and POMDP-style sensing	Task information under partial observability	Observation actions chosen from belief or context	The paper studies a fixed-backbone scheduling setting with limited specialist capacity and replayable references.
DRL-based adaptive sensing and sensor steering	Information gain, coverage, data acquisition utility, or monitoring reward	Learned adaptive policies, often with soft or task-specific constraints	Feasible candidate masks enforce the single specialist operating rule at execution time.
Time series forecasting models	Forecast accuracy after the observation stream is given	Forecasting model for a given observation stream	A fixed forecaster is used as a common evaluator of schedules, not as a separately retrained model per policy.

Deep reinforcement learning has been used for adaptive sensing, informative path planning, and other combinatorial scheduling problems (Murad et al., 2020; Wei and Zheng, 2020; Ogbodo et al., 2026; Pendyala et al., 2024). It is well suited to delayed rewards that depend on state, but hard feasibility remains essential in sensing systems that cannot execute arbitrary actions. The method in this paper therefore uses a masked categorical policy over candidate channel masks. The policy assigns probability only to masks that satisfy the online power and operating rules at the current epoch.

2.3. Forecasting models as schedule evaluators

Modern time-series models provide a direct way to score the prediction consequences of a schedule (Lim et al., 2021; Bai et al., 2018; Liu et al., 2024). In this work, the forecaster is used as a common evaluator of schedules. It is trained on an earlier chronological partition, frozen, and then used to supply reward during policy learning and forecast loss during final evaluation. This choice makes the comparison about the observation pattern induced by each scheduler. The main evaluation uses this common model, and a second forecasting family tests whether the fixed-schedule comparison depends on that evaluator.

2.4. Antarctic blowing-snow monitoring

The application study is motivated by Antarctic automatic weather stations and blowing-snow monitoring. Antarctic AWS networks provide long-term atmospheric and surface observations, but instrument coverage remains uneven in space, season, and measurement type (Wang et al., 2023). Blowing-snow studies use variables whose predictive value changes with storm conditions, near-surface transport, and particle microstructure (Amory, 2020; Sharma et al., 2023; Monrad-Krohn et al., 2026). The simulation maps this domain into a controlled task with a fixed backbone and a single selectable specialist so that learned and fixed specialist strategies can be replayed on identical truth sequences.

3. Problem Formulation

3.1. Forecast objective for sensing system scheduling

Let $x_t \in \mathbb{R}^d$ denote the simulated microclimate state at scheduling epoch t. The state contains meteorological variables and blowing-snow variables, including wind, temperature, humidity, pressure, radiation, snow-surface temperature, particle statistics, and snow mass flux. A channel selection is a binary vector $a_t \in \{0, 1\}^m$ over m logical sensor channels. A logical channel is a data stream that can be scheduled independently.

The reported sensing system uses two channel classes. A required weather backbone remains active because it supplies the minimum context needed by the downstream forecast task. The remaining budget allows at most one specialist channel from a set containing radiation, thermal, particle, and flux measurements. The feasible action set is

$$\begin{aligned}\mathcal{A}_{\text{mask}} = \{a \mid a_{\text{met}} = 1,\ \sum_{i \in \mathcal{S}} a_i \leq 1,\ \sum_i p_i a_i \leq B, \\ \sum_i p_i^{\text{peak}} a_i \mathbf{1}\{a_{t-1,i} = 0\} \leq B^{\text{peak}}\},\end{aligned} \tag{1}$$

where $\mathcal{S}$ is the specialist set. One weather backbone is mandatory, at most one specialist can be active, and startup peaks stay within limits. This geometry makes the comparison identifiable because a fixed design can choose one specialist but cannot observe all specialist streams simultaneously.

Table 2. Notation used for the scheduling formulation and PPO objective.

Symbol	Meaning
x_t	Latent microclimate state at scheduling epoch t
o_t	Scheduler observation built from value/mask histories, runtime and ready-sampling age, previous mask, budget/time state, and supplied warning scores
m^k	Candidate executable channel mask with index k
$\mathcal{I}_t$	Candidate-index set that remains feasible at epoch t after power, startup, and minimum-activation checks
u_t	PPO action index sampled or selected from $\mathcal{I}_t$
$a_t = m^{u_t}$	Executed binary channel mask
$\mathcal{L}_{\text{step}}$	Ordinary multi-horizon forecast loss
$\mathcal{M}_{\text{macro}}$	Macro score normalized by fixed-schedule references across event regimes
S_t	Switching or minimum-activation penalty term used during policy learning

For PPO training and evaluation, feasible channel masks are represented as a finite candidate set $\{m^1, \ldots, m^K\} \subseteq \mathcal{A}_{\text{mask}}$. At epoch t, online power, startup, and minimum-activation checks define the currently feasible index set $\mathcal{I}_t \subseteq \{1, \ldots, K\}$. The policy samples or selects an action index $u_t \in \mathcal{I}_t$, and the executed channel selection is $a_t = m^{u_t}$. This notation separates the PPO action from the binary mask applied to the sensing system.

3.2. *When changing specialists can help*

The single specialist setting is a controlled instance of a broader problem with limited specialist capacity. Let $\mathcal{B}_0$ denote mandatory backbone channels and let $\mathcal{S}$ contain K specialist channels. At each step the scheduler may select at most $r < K$ specialists. Let $c \in C$ denote latent forecast regimes with positive evaluation weights ρ_c, and let $S_c^\star \subseteq \mathcal{S}$ be the best specialist subset for regime c, with $|S_c^\star| \leq r$.

A dynamic scheduling advantage can arise when no fixed subset $S_0 \subseteq \mathcal{S}$ with $|S_0| \leq r$ matches $S_c^\star$ for all regimes with positive evaluation weight, and serving a regime with a mismatched specialist subset has positive excess forecast loss. The regime label need not be observed directly at deployment time. The condition states when observable regime evidence can make dynamic specialist allocation valuable.

Specialist channel

	Radiation	**Thermal**	**Particle**	**Flux**
Particle regime	mismatch	mismatch	matched	partial
Flux regime	mismatch	partial	mismatch	matched
Thermal regime	partial	matched	mismatch	mismatch

Qualitative guide. Green cells indicate the low-loss specialist for a regime; grey cells indicate a fixed-specialist mismatch.

Proposition 1 (Dynamic value with one selectable specialist)**.** *Consider a scheduling problem with one selectable specialist and common startup and minimum-activation rules. Suppose at least two regimes with positive evaluation weight* *have incompatible best specialists, and their intervals are long enough for a regime-informed policy* π^{lab} *to change specialists feasibly. If, after transition losses are included, the normalized regime loss of every fixed specialist is no lower than that of* π^{lab} *in every regime and exceeds it by at least* $\Delta_c > 0$ *in each positive-weight regime that it mismatches, then every fixed specialist has strictly higher macro forecast loss than* π^{lab}.

A proof is given in Appendix A.1. The proposition is a sufficient condition for adaptive value under the same temporal action rules; it imposes no upper bound on the performance of an implemented policy. In the reported simulation, particle, flux, and thermal regimes favor different specialists. The exact-label replay in Table 5 is a finite, validation-mapped diagnostic that receives the held-out simulator subtype. It measures the value of direct regime information under that mapping and need not realize π^{lab} or minimize forecast loss.

Verification of the incompatibility condition. The incompatibility condition in Proposition 1 is built into the controlled action geometry and observation model: particle, flux, and thermal intervals expose different target variables most reliably through the laser, FC4, and surface-infrared channels. The held-out regime decomposition in Table 9 then tests whether the learned scheduler uses this opportunity and improves the equal-weight macro objective, without assuming that every individual regime must improve in every seed.

3.3. Forecast loss and macro scoring across regimes

The fixed evaluation forecaster maps a window of sample-and-hold values and observation masks to forecasts of selected target variables over future horizons. Let $\hat{y}_{t+k}$ denote the prediction at horizon k and y_{t+k} the corresponding truth. Let c_t

denote the offline evaluation regime at epoch t. With horizon weights w_k, target weights α_{j,c_t}, and target scales σ_j, the per-step forecast loss is

$$\mathcal{L}_{\text{step}}(t) = \frac{\sum_{k=1}^{H} \sum_{j\in\mathcal{Y}} w_k \alpha_{j,c_t} \left|\hat{y}_{t+k,j} - y_{t+k,j}\right| / \sigma_j}{\sum_{k=1}^{H} \sum_{j\in\mathcal{Y}} w_k \alpha_{j,c_t}}. \tag{2}$$

The reported experiment uses uniform horizon weights and the fixed target weights in Table D.15. Event labels select these predeclared weights only in the common offline evaluator; they are not policy inputs. All schedules are scored with the same target scales, weights, and labels.

Average forecast loss can favor the event type that is easiest or most frequent. The main reported score therefore uses macro averaging across event types. Let $c \in \{\text{particle}, \text{flux}, \text{thermal}\}$ index the event types, and let $\mathcal{T}_c$ be the evaluated final-test epochs labelled as type c by the common offline simulator record. For policy π, define

$$L_c(\pi) = \frac{1}{|\mathcal{T}_c|} \sum_{t\in\mathcal{T}_c} \mathcal{L}_{\text{step}}^{(c)}(t; \pi). \tag{3}$$

The fixed-schedule reference denominator s_c is the median event type loss across the feasible fixed-schedule candidates evaluated on the validation partition. The macro score normalized by fixed-schedule references is

$$\mathcal{M}_{\text{macro}}(\pi) = \frac{1}{3} \sum_{c\in\{\text{particle},\text{flux},\text{thermal}\}} \frac{L_c(\pi)}{s_c}. \tag{4}$$

Lower values are better. The score asks whether a scheduler improves balanced forecasting across event regimes relative to the difficulty of those regimes for fixed schedules.

3.4. Why forecast loss is not an AoI or covariance proxy

AoI and covariance trace remain valid scheduling objectives, but they optimize different quantities. Let π be a feasible scheduling policy. Its forecast value is

$$J_{\text{fcst}}(\pi) = \mathbb{E}_{\pi}\left[-\sum_{t=0}^{T-1} \gamma^t \mathcal{L}_{\text{step}}(t)\right]. \tag{5}$$

AoI and covariance policies optimize

$$J_{\text{AoI}}(\pi) = -\mathbb{E}_{\pi}\left[\sum_{t=0}^{T-1} \sum_{j \in \mathcal{Y}} \text{age}_{t,j}\right], \qquad J_{\text{cov}}(\pi) = -\mathbb{E}_{\pi}\left[\sum_{t} \text{tr}(P_t)\right]. \tag{6}$$

These objectives can rank the same schedules differently when a delayed specialist is valuable before event intervals.

Proposition 2 (Non-equivalence of forecast loss reward)**.** *Forecast-loss reward, estimator covariance trace, and per-variable AoI can induce different orderings over feasible activation-delay schedules. In particular, there exist two feasible policies π_1 and π_2 such that π_1 has lower AoI or lower covariance trace, while π_2 has strictly lower forecast loss.*

The proof in Appendix A.2 uses a two-variable linear-Gaussian construction in which the event sensitive variable has a large coefficient $M \gg c$ relative to the background variable. The construction shows that when event-period forecast loss is dominated by the event sensitive variable, a policy that delays background refreshes to activate the event sensor can have strictly lower forecast loss despite higher AoI and higher covariance trace. AoI and covariance objectives are therefore not substitutes for a direct forecast loss comparison when event sensitive channels exist; they are included as rule-based schedules to provide non-learned dynamic comparators, not as equivalent optimization targets.

The proposition establishes non-equivalence of the objectives, not guaranteed superiority of one trained policy in every finite experiment. The matched reward controls in Sections 4.5 and 6.3 test the empirical separation while holding the PPO architecture and feasibility interface fixed.

4. PD-PPO Scheduler and Evaluation Protocol

4.1. State, action, and feasibility masking

PD-PPO is the masked policy used for the forecast objective in Equations (2) and (4). At each epoch, the policy receives recent sample-and-hold values and observation masks, channel ready-sampling age, sensor runtime state, the previous action mask, budget and time features, and supplied event-warning scores. In the reported implementation, the action space is the finite set of candidate masks in Table 3. Online checks remove masks that would violate power, startup, or minimum-activation constraints. The actor assigns logits to the remaining masks and samples, or greedily selects, one executable mask.

Feasibility masking is part of the deployed policy. The learned controller is optimized directly over executable actions; no fallback repair follows selection. Comparison policies are passed through the same feasible-action interface when the comparison requires matched execution rules.

Table 3 is the executable action surface used by both the learned scheduler and matched references. Every mask keeps the weather backbone active and differs only in the selectable specialist sensor made available for forecasting. This paper therefore tests the resulting fixed-backbone, one-specialist operating regime.

Table 3. Executable sensor choices in the reported one-specialist simulation.

Id	Executed channels	Power	Interpretation	Fixed selections
0	Weather backbone	0.25	Backbone only	0/24
1	Weather backbone + Radiometer	0.33	Radiation context	0/24
2	Weather backbone + Thermo-hygrometer	0.43	Non-event weather context	0/24
3	Weather backbone + Surface infrared sensor	0.74	Thermal-event specialist	9/24
4	Weather backbone + Laser disdrometer	0.74	Particle-event specialist	2/24
5	Weather backbone + FC4 flux sensor	0.74	Flux-event specialist	13/24

Every action includes the weather backbone. Fixed selections are validation-selected fixed schedules over the 24 final test seeds.

4.2. Partial observations and scheduler input

The environment maintains a sample-and-hold state rather than a model-based Kalman estimate. Let $y_{t,i,j}$ be a valid measurement of variable j from a ready channel i, and let $R_{i,j}$ be its configured noise variance. When one or more ready channels observe the same scalar variable, the current value is the inverse-variance weighted mean

$$\widetilde{x}_{t,j} = \frac{\sum_{i \in O_{t,j}} R_{i,j}^{-1} y_{t,i,j}}{\sum_{i \in O_{t,j}} R_{i,j}^{-1}}. \tag{7}$$

Wind direction is combined by the corresponding weighted circular mean. If $O_{t,j}$ is empty, the last value is retained, $\widetilde{x}_{t,j} = \widetilde{x}_{t-1,j}$. A binary mask records whether each value was observed at epoch t. The scheduler and fixed forecaster receive the last 20 value and mask vectors, so they can distinguish a new measurement from a carried value.

The scheduler state also contains each channel's off/warming/ready mode, remaining warm-up time, and normalized age since it last reached a ready sampling opportunity. This channel-level quantity is an AoI-inspired runtime proxy: it resets when an active channel is ready to sample, even if a stochastic measurement attempt yields no valid variable. The previous executed mask, current power ratio, diurnal phase, and a compact summary derived from the partial history complete the physical state. The main policy does not receive an estimated covariance matrix. Covariance is retained in Proposition 2 as a competing objective, and a diagonal predict–update proxy is used only in the matched uncertainty-reward control.

To summarize recent physical evidence without using event labels, the policy also receives ten deterministic history indicators. Let $\bar{x}_{t,j}$ be the sample-and-hold value normalized by forecaster-fitting statistics and clipped to $[-5, 5]$, let $\Delta_{16}\bar{x}_{t,j} = \bar{x}_{t,j} - \bar{x}_{t-15,j}$, and let $C_{16}(\mathcal{J})$ be the largest 16-step observation fraction among variables in $\mathcal{J}$. Three bounded indicators are

$$b_t^{\text{part}} = \tanh(0.45\bar{v}_{p,t} + 0.25\bar{d}_{p,t} + 0.15\bar{f}_t), \tag{8}$$

$$b_t^{\text{flux}} = \tanh(0.55\bar{f}_t + 0.25\bar{u}_t + 0.10\Delta_{16}\bar{f}_t), \tag{9}$$

$$b_t^{\text{therm}} = \tanh\big(0.45(\bar{T}_{a,t} - \bar{T}_{s,t}) - 0.10\Delta_{16}\bar{T}_{s,t}\big)\,. \tag{10}$$

Here u, f, v_p, d_p, T_a, and T_s denote wind speed, mass flux, particle velocity, particle diameter, air temperature, and surface temperature. The three indicators are concatenated with current normalized wind speed, mass flux, particle velocity, the air–surface temperature difference, and the particle, flux, and thermal coverage terms. Every component is computed from the same carried values and masks already available online.

Four exogenous warning scores provide particle, flux, thermal, and aggregate event context. In the controlled simulator, each subtype score is generated from the corresponding event process by adding a leading ramp and Gaussian noise. These scores model an upstream warning input available at decision time; they are not the exact event labels used for common offline weighting and grouping. Their construction and parameters are reported in Appendix D.

The actor uses the aggregate warning $q_t^{\max}$ to blend two observation encoders,

$$h_t = (1 - g_t) f_0(o_t) + g_t f_1(o_t), \qquad g_t = \mathrm{sigmoid}(\alpha q_t^{\max} + \beta), \tag{11}$$

where α, β, and both encoders are learned. A candidate mask m^u is represented by the sum of learned embeddings of its active channels. Its actor logit is the scaled inner product between this mask embedding and h_t, plus learned mask biases; infeasible logits are then set to a large negative value before categorical sampling. The critic receives o_t and $q_t^{\max}$ through a separate multilayer perceptron. Thus, warning context modulates the policy without supplying the exact event subtype.

4.3. *Fixed evaluation forecaster and training reward*

The evaluation forecaster is trained before policy learning and then held fixed. It receives the 20-step sample-and-hold value history and its observation masks; Ready-sampling age, runtime, warning scores, and the history indicators above are scheduler inputs rather than forecaster inputs. During policy training, it returns multi-step forecast loss for the trajectory induced by the current schedule. During final evaluation, the same forecaster family scores all schedules on held-out windows. The comparison therefore concerns the executed observation pattern, not retraining a different forecaster for each policy.

Table 4 separates quantities available to the online scheduler from labels and future values used only for training, supplementary analysis, or final scoring. This

Algorithm 1 Masked PD-PPO training

Fit the evaluation forecaster on the forecaster fitting partition.

Replay fixed candidate masks on the calibration partition; compute and freeze the subtype loss scales d_c.

for each policy training rollout **do**

  Build o_t from value/mask histories, channel runtime and ready-sampling age, previous mask, budget/time state, and supplied warning scores.

  Enumerate currently feasible mask indices $\mathcal{I}_t$.

  Sample $u_t \sim \pi_\theta(\cdot \mid o_t, \mathcal{I}_t)$ and execute mask $a_t = m^{u_t}$.

  Score the induced partial observation window with the fixed forecaster and form the reward in Equation (12).

end for

Update PPO with the clipped objective, value loss, entropy term, and the labelled training auxiliaries.

Freeze the learned checkpoint before final test evaluation.

distinction prevents the event-label reference from being confused with the learned controller.

The policy reward is the negative of forecast loss plus operational penalties.

$$r_t = -\left[\lambda_{\text{fcst}}\mathcal{L}_{\text{train}}(t) + \lambda_{\text{sw}}S_t + \lambda_{\text{viol}}V_t\right], \tag{12}$$

where $\mathcal{L}_{\text{train}}$ is the forecaster loss used during policy learning, S_t is switching cost, and V_t collects feasibility violations. The reported evaluation separates the mean forecast loss from the macro score in Equations (2) and (4).

The training loss aligns the policy update with the balanced event objective. For training-partition subtype c_t, it is

$$\mathcal{L}_{\text{train}}(t) = \omega_{c_t}\frac{\mathcal{L}_{\text{FW}}(t)}{d_{c_t}}, \tag{13}$$

where d_c is the median loss of the fixed candidate masks on the calibration partition and ω_c is a fixed event weight. The particle, flux, and thermal weights are 1.55, 0.95, and 1.65; calm epochs use unit weight and the default fixed-candidate scale. These constants are computed before policy optimization and remain fixed during training and final evaluation. The subtype label used in Equation (13) is available only while training on the generated sequence. Final policy execution requires the observable warning scores but neither c_t nor a reward value.

Policy learning uses two training-only supervised signals in addition to the forecast reward. A simulator subtype label within a 16-step look-ahead window selects a predefined feasible guide mask: the weather backbone is paired with the thermo-hygrometer in calm periods, the disdrometer for particle events, the flux sensor for transport events, and the infrared thermometer for thermal events. This guide supplies behavior-cloning targets, while the same four-class label supervises an auxiliary context head. Both signals are formed only on the policy-training

Table 4. Online availability of scheduler inputs and diagnostic labels.

Quantity	Source	Online	Use in the protocol
Sample-and-hold values	Valid measurements with last-value carry-forward	Yes	Policy and fixed-forecaster input
Observation mask and ready-sampling age	Runtime state and valid-observation log	Yes	Policy input; also used by conventional rules
History indicators	Last 16 carried values and observation masks	Yes	Policy input; no event label required
Previous mask and warm-up state	Scheduler runtime	Yes	Feasibility mask and switching/minimum-activation accounting
Synthetic event-warning scores	Event-process-derived noisy lead signals supplied as context	Yes	Policy input under the simulator assumption
Event-type labels	Simulator truth sequence	No	Training guide, auxiliary labels, common evaluator weights, and diagnostic grouping
Future target values	Simulator truth sequence	No	Fixed-forecaster fitting, reward scoring, and final evaluation
Loss normalizers and selected fixed schedule	Calibration partition	No	Training loss scaling, reference selection, and metric normalization

partition. The actor and critic themselves receive the supplied warning score, rather than the exact subtype label, during policy rollouts and final evaluation. The guide first provides a 10,000-step supervised warm start; after PPO begins, its contribution is weighted by the policy advantage as in Equation (18).

The evaluation forecaster is a residual temporal convolutional network (Bai et al., 2018) with three levels, 64 channels per level, kernel size 3, and dropout rate 0.05. It is fitted on the forecaster training partition before policy learning. Input and target scales are fitted before policy updates, and the architecture is held fixed throughout policy learning and final evaluation. No forecaster weights are updated after the forecaster fitting partition ends.

The fitting mixture contains 78 rollouts and 53,274 rollout epochs, all drawn from that partition. Full observation contributes 18 rollouts. Feasible static, round-robin, AoI-priority, and random schedules contribute six each, as do each of the six fixed candidate masks. Each rollout contains 683 epochs; for each policy specification, three starts are event centered and three are sampled uniformly from the fitting partition. At reset, all channels are off, the 20-epoch value history is initialized to the fitting-partition mean, and its observation masks are zero. This mixture exposes the forecaster to both dense and schedule-induced partial histories before it is frozen; it does not contain validation or final-test observations.

4.4. PPO objective

Let o_t denote the scheduler observation, u_t the selected candidate mask index, $a_t = m^{u_t}$ the executed channel mask, and $\hat{A}_t$ the advantage estimate. The policy defines a masked categorical distribution,

$$\pi_\theta(u \mid o_t) = \frac{\exp z_\theta(o_t, u)}{\sum_{v \in \mathcal{I}_t} \exp z_\theta(o_t, v)} \mathbf{1}\{u \in \mathcal{I}_t\}, \tag{14}$$

where $z_\theta(o_t, u)$ is the actor logit for candidate mask m^u and $\mathcal{I}_t$ is the online feasible index set. PPO updates this distribution with the clipped surrogate

$$\mathcal{L}_{\text{clip}}(\theta) = -\mathbb{E}_t \left[\min \left(\rho_t(\theta)\hat{A}_t,\ \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\,\hat{A}_t\right)\right], \tag{15}$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(u_t \mid o_t)}{\pi_{\theta_{\text{old}}}(u_t \mid o_t)}. \tag{16}$$

The reported PD-PPO loss contains the PPO surrogate, value loss, entropy regularization, and two auxiliary terms tied to the forecast training setup.

$$\mathcal{L}_{\text{PD-PPO}} = \mathcal{L}_{\text{clip}} + c_v \mathcal{L}_V - c_H \mathcal{H}(\pi_\theta) + \beta_{\text{BC}} \mathcal{L}_{\text{BC}} + \beta_{\text{event}} \mathcal{L}_{\text{event}}. \tag{17}$$

For labelled training states, let g_t denote the feasible subtype guide mask defined above. The advantage-weighted behavior-cloning term is

$$\mathcal{L}_{\text{BC}} = -\frac{\sum_t \ell_t [\hat{A}_t]_+ \log \pi_\theta(g_t \mid o_t)}{\sum_t \ell_t [\hat{A}_t]_+ + \varepsilon}, \tag{18}$$

where $\ell_t \in \{0, 1\}$ marks states with a valid guide. The event-context auxiliary signal is a cross-entropy loss on a separate classifier head. Let h_t be the policy representation, $q_\theta(c \mid h_t)$ the auxiliary probability of context class c, and $c_t \in \{\text{calm}, \text{particle}, \text{flux}, \text{thermal}\}$ the simulator-derived label used only during policy training. With labelled state indicator η_t, the term is

$$\mathcal{L}_{\text{event}} = -\frac{\sum_t \eta_t \log q_\theta(c_t \mid h_t)}{\sum_t \eta_t + \varepsilon}. \tag{19}$$

Full protocol and training hyperparameters are listed in Appendix C.

4.5. Matched objective and learner controls

Two controls retain the masked PPO architecture, online observation, six candidate masks, feasibility checks, chronological partitions, event weights, training guide, and auxiliary context loss, but replace the scalar forecast loss in the reward. For target variable set $\mathcal{J}$ and history length H, the AoI control uses

$$\mathcal{L}_{\mathrm{AoI}}(t) = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \min\left(\frac{a_{t,j}}{H-1}, 1\right), \tag{20}$$

where $a_{t,j}$ is the number of epochs since the latest valid observation of variable j in the input history. A variable with no valid observation in the history receives unit loss.

The uncertainty control maintains a diagonal variance only for its reward. For each variable, it applies

$$P^-_{t,j} = \min(P_{t-1,j} + Q_j, P_{\max}), \qquad P_{t,j} = \left[(P^-_{t,j})^{-1} + R^{-1}_{t,j}\right]^{-1} \tag{21}$$

when a valid observation is available, and $P_{t,j} = P^-_{t,j}$ otherwise. $R_{t,j}$ is the inverse-variance fused measurement variance from Equation (7). Its bounded reward proxy is

$$\mathcal{L}_{\mathrm{unc}}(t) = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \frac{P_{t,j}}{1 + P_{t,j}}. \tag{22}$$

These controls isolate the reward proxy conditional on the same event-weighted training protocol; they do not remove the training-only guide or auxiliary label.

The learned-policy control is a masked dueling Double-DQN. It receives the same online state, forecast reward, candidate masks, feasible-action mask, training partition, and final windows as PD-PPO. Online and target networks implement the Double-DQN target over feasible next actions, with three-step returns and experience replay. This control does not use the PPO behavior-cloning warm start or auxiliary classifier. It therefore tests a second value-based learner on the same decision surface rather than serving as a one-component PPO ablation. Its complete settings are listed in Table C.12.

4.6. Chronological separation and comparison references

Each generated truth sequence is split into chronological partitions for forecaster fitting, policy training, calibration and fixed-schedule selection, and final testing. The calibration partition fixes the subtype loss scales and the fixed comparison mask. Final test windows are not used for forecaster fitting, policy updates, imitation-label generation, fixed-schedule selection, or normalizer fitting.

The chronological split is summarized in Appendix C.

The comparison families serve different roles. Validation-selected fixed schedules test whether adaptation is needed at all. Conventional rule-based references test simple non-learned adaptation under the same operating rules. The one-step forecast-greedy reference chooses the currently best feasible mask under the fixed forecaster and therefore probes whether the learned policy has sequential value. The warning-score rule maps the same supplied synthetic warning scores to masks selected on validation data. It provides a strong memoryless context reference and does not define the primary comparison. The exact-label diagnostic receives the held-out simulator subtype and measures how useful direct regime information can be; it is not a deployable learned controller or a guaranteed optimizer.

Table 5 fixes the role of each reference before the results section.

Exact decision rules, validation choices, tie breaks, and information privileges for these references are listed in Appendix E.

5. Simulation and Evaluation Setup

5.1. Simulated sensing system

The evaluation uses a controlled cold-region sensing system simulation motivated by Antarctic automatic weather stations (AWSs). The final experiment

Table 5. Evaluation references and matched controls.

Schedule family	Role in the protocol	Interpretation
Validation-selected fixed schedule	Select one feasible constant schedule before final testing.	Primary fixed reference.
AoI, round-robin, and random schedules	Use feasible cycling, AoI priority, or random rules.	Non-learned dynamic references.
Masked Double-DQN	Learn on the same observation, forecast reward, candidate masks, and feasibility interface.	Value-based learned-policy control.
One-step forecast-greedy reference	Choose the currently best feasible mask under the fixed forecaster.	Privileged myopic diagnostic for sequential value.
Warning-score rule	Map the supplied synthetic warning scores to validation-selected masks.	Strong memoryless context reference.
Exact-label diagnostic	Map held-out simulator event subtypes to validation-selected masks.	Privileged information diagnostic; not the learned controller.
Matched reward controls	Retrain the same masked PPO with AoI or diagonal-uncertainty reward.	Isolates the scalar reward proxy under the shared training protocol.

uses six logical channels, consisting of one required weather backbone and five candidate specialists. Table 6 lists the channels and their normalized scheduling costs. A platform rendering is retained in Appendix B as physical motivation, while the channel table defines the schedulable abstraction used in the experiments.

The fixed backbone with one selectable specialist is a controlled abstraction of a practical choice: keeping basic weather context available while allocating the remaining active slot among specialist instruments whose forecast value changes across blowing-snow regimes.

5.2. Truth-sequence generation

Each seed generates a multivariate truth sequence with wind, temperature, humidity, pressure, radiation, snow-surface temperature, particle statistics, and snow mass flux. The generator is anchored to Antarctic AWS and blowing-snow statistics (Wang et al., 2023; Amory, 2020; Sharma et al., 2023; Monrad-Krohn et al., 2026). Blowing-snow events are produced by a semi-Markov process coupled to wind exceedance.

The final simulation contains particle, flux, and thermal event types. Each event type changes a different part of the latent process and is observed most reliably by a different specialist channel. The laser disdrometer is the particle specialist, the FC4 flux sensor is the flux specialist, and the surface infrared temperature sensor is the thermal specialist. Event balance, event weighted targets, and the single specialist action rule make all three regimes matter in the forecast score. One fixed specialist can be effective on average, but no single specialist covers the full forecast objective.

The scheduler is also supplied with noisy leading warning scores for the three event types. These synthetic scores represent an upstream warning service in the

Table 6. Logical channels and normalized scheduling costs.

Idx	Logical channel	Main observed variables	p_i	p_i^{peak}	Warm-up
1	Weather backbone	wind, air temperature, humidity, pressure	0.25	0.30	0
2	Basic radiometer	solar radiation	0.08	0.10	0
3	Shielded thermo-hygrometer	air temperature, humidity	0.18	0.22	0
4	Surface infrared temperature sensor	snow-surface temperature, thermal-event evidence	0.49	0.62	1
5	Laser disdrometer	particle diameter, particle velocity, particle microstructure evidence	0.49	0.62	2
6	FC4 blowing-snow flux sensor	snow mass flux, flux-event evidence	0.49	0.62	1

The backbone is intentionally treated as required context. The scheduling decision is the specialist choice. Surface infrared temperature, laser disdrometer, and FC4 flux are high-value specialists for thermal, particle, and flux event regimes, respectively; the low-cost radiometer and thermo-hygrometer channels are retained as plausible non-event alternatives. Event type labels are simulator-derived training and offline analysis labels, not direct sensor observations.

controlled experiment; their construction, and the distinction from exact event labels, are specified in Appendix D.

Generator checks cover meteorological structure and blowing-snow behavior, including wind autocorrelation, event frequency, scalar-distribution agreement with station anchors, event durations, flux–wind scaling, particle correlation, and wind-speed spectra (Aloni et al., 2025). All eight criteria pass their prespecified acceptance rules; the full check table and generator-validation figure are reported in Appendix F. These checks document the simulation construction and remain separate from final test policy selection.

5.3. Metrics, seeds, and statistical summaries

The primary evaluation uses eight 512-epoch windows selected inside the final partition by a prespecified subtype-balanced, transport-rich rule. The rule uses event coverage and transport intensity from the generated truth sequence, not any policy loss, and is applied identically to every schedule. Mean forecast loss averages Equation (2) over the resulting 4096 epochs. The primary balanced score is the macro score normalized by fixed schedules in Equation (4). Metric normalizers are computed from feasible fixed-schedule candidates on the validation partition, so the score is specific to this simulation family. The fixed evaluation forecaster architecture and training setup are described in Section 4.

As a coverage sensitivity, the same frozen policies and validation-selected fixed schedules are also replayed continuously over every final-partition epoch with a complete eight-step future target. This gives 5,242 scored epochs per seed, covering $[64750, 69992)$; only the last eight epochs are excluded because their future targets are incomplete. The policies, fixed masks, and validation-fitted normalizers are unchanged, and no schedule is reselected from this replay.

The reported main aggregate uses 24 evaluation seeds, numbered 117–140. For each seed, the same truth sequence is used for PD-PPO, fixed-schedule references, conventional rule-based schedules, and supplementary references. The selected fixed schedule is listed before final testing and then replayed as a constant feasible schedule without fallback rotations.

Seeds 117 and 118 were used for a bounded architecture choice between the base warning-gated actor and a variant with an additional engineered-context encoder. The base actor was selected before the remaining seeds were run. The unchanged post-pilot subset, seeds 119–140, retains positive primary macro margins in 22/22 seeds (mean 0.0838; 95% interval [0.0707, 0.0967]). Results report this replication check alongside the full 24-seed aggregate. The scenario parameters, target weights, reference definitions, and reported training settings were frozen for this expansion.

The main protocol hyperparameters are listed in Appendix C. Per-seed selected fixed schedules and validation scores are listed in Appendix G.

Forecaster sensitivity is tested by fitting a multi-output ridge model on the same forecaster-fitting partition for each seed. The ridge model selects its own fixed specialist using validation windows and then rescores the unchanged final PD-PPO and fixed trajectories. No scheduling policy, action trace, or forecasting model is updated from final results. Exact subtype labels select the same predeclared target weights for every schedule and group completed losses for the macro score.

Seed statistical summaries use paired differences because each scheduler and reference is replayed on the same truth sequence. For a seed s, a positive margin means reference loss minus PD-PPO loss. Alongside mean margins and win counts, the results report medians and percentile bootstrap 95% confidence intervals over seeds.

5.4. *Action-trace checks*

Action-trace checks are computed from held-out action traces. They report mask entropy, transition entropy, event–action mutual information, and sensor use separation by regime. The audit flags a trace as fixed like when one mask explains at least 95% of actions, and as simple cycle like when the top three masks explain at least 85% of actions and the best periodic match up to lag 64 reaches 90%, unless there is sufficient state dependence. These checks support interpretation of the learned schedule; the primary performance claim remains the paired forecast loss comparison.

6. Results

The evaluation examines held-out forecasting performance, the schedules that produce it, and three possible alternative explanations: the choice of reward, the PPO learner, and the fixed forecasting model used for evaluation.

6.1. *Held-out forecasting performance*

Table 7 gives the paired results for all 24 final seeds. PD-PPO has lower mean forecast loss and lower equal-weight regime score than the validation-selected fixed schedule in every seed. Mean forecast loss decreases from 1.5592 to 1.4012, a relative reduction of 10.1%. The macro score decreases from 1.0206 to 0.9404, a relative reduction of 7.9%. The mean fixed-schedule margins are +0.1580 for mean loss and +0.0801 for the macro score, with bootstrap intervals that remain above zero. An exact one-sided sign test for 24 positive paired macro differences gives $p = 5.96 \times 10^{-8}$.

The first two seeds were used to verify the frozen protocol before expansion. The unchanged post-pilot seeds 119–140 independently retain 22/22 macro wins, with a mean margin of +0.0838 and a 95% bootstrap interval of [+0.0707, +0.0967]. The replication shows that the result persists beyond the pilot pair. PD-PPO also improves on the AoI-priority, round-robin, and random feasible schedules in 24/24 seeds for both reported losses.

Table 7. Held-out schedule comparisons for PD-PPO. Positive margins mean that the reference has higher loss than PD-PPO.

Reference	Mean loss: wins; margin [95% CI]	Macro: wins; margin [95% CI]
Validation-selected fixed schedule	24/24; +0.1580 [+0.1161, +0.2052]	24/24; +0.0801 [+0.0673, +0.0931]
AoI-priority schedule	24/24; +0.2572 [+0.2285, +0.2872]	24/24; +0.1602 [+0.1385, +0.1838]
Round robin	24/24; +0.2519 [+0.2206, +0.2836]	24/24; +0.1609 [+0.1380, +0.1854]
Random feasible schedule	24/24; +0.2620 [+0.2311, +0.2950]	24/24; +0.1624 [+0.1391, +0.1869]
One-step forecast greedy	24/24; +0.2690 [+0.2347, +0.3040]	24/24; +0.1790 [+0.1507, +0.2097]
Warning-score rule	10/24; -0.0034 [-0.0093, +0.0018]	11/24; +0.0012 [-0.0043, +0.0076]
Exact-label replay	10/24; -0.0040 [-0.0100, +0.0009]	12/24; +0.0018 [-0.0039, +0.0081]

The fixed, AoI, round-robin, and random schedules are deployable references under the common feasibility interface. Forecast greedy can inspect one-step future loss; exact-label replay receives held-out simulator labels; the warning-score rule receives supplied synthetic warning scores.

The primary windows deliberately balance particle, flux, and thermal regimes. Table 8 tests whether that sampling rule drives the result by replaying the same frozen schedules continuously over all 5,242 scoreable epochs of each final partition. PD-PPO again has lower mean and macro loss in 24/24 seeds. The mean margins are +0.1247 for ordinary forecast loss and +0.0793 for the macro score, and both bootstrap intervals remain above zero. The result therefore extends beyond the transport-rich windows used for the primary balanced comparison.

Table 8. Sensitivity to final-partition coverage. Positive margins mean that the validation-selected fixed schedule has higher loss than PD-PPO.

Evaluation scope	Epochs per seed	Mean loss: PD-PPO / fixed	Mean-loss wins; margin [95% CI]	Macro: PD-PPO / fixed	Macro wins; margin [95% CI]
Prespecified subtype-balanced windows	4,096	1.4012 / 1.5592	24/24; +0.1580 [+0.1161, +0.2052]	0.9404 / 1.0206	24/24; +0.0801 [+0.0673, +0.0931]
Complete scoreable final partition	5,242	0.6632 / 0.7880	24/24; +0.1247 [+0.0898, +0.1641]	0.4301 / 0.5093	24/24; +0.0793 [+0.0642, +0.0951]

The continuous replay uses the same frozen checkpoints, validation-selected masks, and validation-fitted subtype normalizers as the primary evaluation. The last eight rows of each final partition are excluded because an eight-step future target is unavailable; no policy or reference is reselected.

6.2. Regime allocation and action traces

The aggregate gain is distributed across all three forecast regimes (Table 9). Mean margins are positive for particle, flux, and thermal windows, and each bootstrap interval remains above zero. The seed-level counts differ by regime: PD-PPO is better in 20/24 particle, 13/24 flux, and 17/24 thermal comparisons. Equal weighting combines the three incompatible measurement requirements and yields a positive macro margin in 24/24 seeds.

Figure 3 shows the corresponding decisions. The laser disdrometer is active for 99.3% of particle-regime steps, FC4 for 97.8% of flux-regime steps, and the

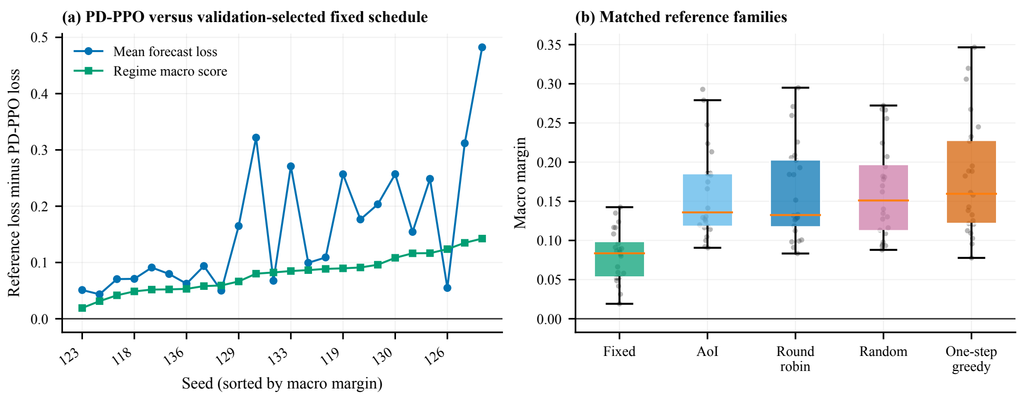

Figure 2. Held-out schedule comparisons over 24 seeds. Panel A shows the mean-loss and regime-macro margins against each seed's validation-selected fixed schedule, sorted by macro margin. Panel B compares the macro margins for the matched fixed, rule-based, and one-step greedy references. Positive values favor PD-PPO; all points in both panels are positive.

Table 9. Validation-normalized loss by forecast regime for PD-PPO and the validation-selected fixed schedule. Positive margins favor PD-PPO.

Regime	PD-PPO	Fixed	Positive seeds	Margin [95% CI]
Particle regime	1.0394	1.0678	20/24	+0.0284 [+0.0152, +0.0459]
Flux regime	0.7535	0.8706	13/24	+0.1171 [+0.0557, +0.1786]
Thermal regime	1.0285	1.1233	17/24	+0.0948 [+0.0492, +0.1422]
Equal-weight macro	0.9404	1.0206	24/24	+0.0801 [+0.0673, +0.0931]

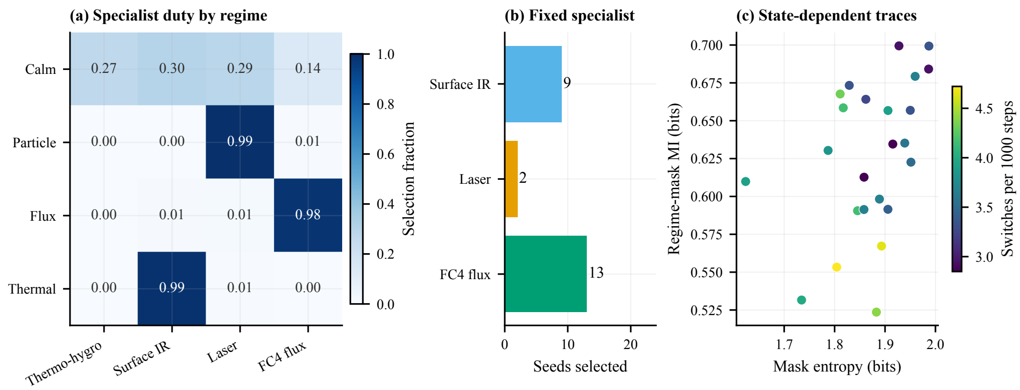

Figure 3. Specialist allocation under the learned and fixed schedules. Panel A reports the mean specialist duty in each forecast regime. Panel B counts the specialist chosen by validation for the fixed reference. Panel C shows the entropy and regime–mask mutual information of each learned action trace; color denotes its switching rate. PD-PPO changes specialists with regime, whereas each fixed reference retains one specialist throughout final evaluation.

surface infrared channel for 99.1% of thermal-regime steps. Calm periods use a mixture of the four selectable specialists. Validation instead selects FC4 in 13 seeds, surface infrared in nine, and the laser disdrometer in two, after which the selected mask remains fixed on final windows.

Every PD-PPO trace passes the prespecified state-dependence, diversity, non-fixed, and non-cyclic checks. Mean mask entropy is 1.872 bits and mean regime–mask mutual information is 0.626 bits. The mean switching rate is 0.00368 per step. The weather backbone remains active by construction, while the policy does not select the low-value basic-radiometer action in these final windows. At least three of the four useful specialists have intermediate overall duty in every seed; all four do so in 23/24 seeds. There are no warm-up aborts. These traces attribute the gain to sparse, regime-dependent reallocation rather than to a constant mask or rapid periodic rotation.

6.3. Objective and learner controls

The matched reward controls replace only the scalar forecast loss while retaining the PPO architecture, observation, feasible masks, training guide, auxiliary context task, chronological partitions, and common final evaluator. Their differences from forecast-reward PD-PPO are small (Table 10). Against the AoI-reward policy, the mean macro difference is +0.0010 with 13/24 wins; against the diagonal-uncertainty-reward policy, it is +0.0008 with 12/24 wins. Both intervals include zero, as do the mean-loss intervals. All three reward variants nevertheless beat the fixed schedule on the macro score in 24/24 seeds and satisfy the execution check in every seed.

The matched Double-DQN uses the same online state, forecast reward, feasible masks, training partition, and final evaluator. PD-PPO has lower macro loss in 24/24 seeds and lower mean loss in 23/24, with mean DQN-minus-PD-PPO differences of +0.0697 and +0.1408, respectively. Double-DQN beats the fixed schedule in 12/24 macro comparisons and passes the full action-trace gate in 21/24 seeds, although it records no warm-up aborts. The result supports the complete masked PPO training design on this decision surface; it is not a single-component comparison because Double-DQN does not use the PPO behavior-cloning warm start or auxiliary classifier.

6.4. Myopic and event-information references

One-step forecast greedy is a privileged myopic reference: at each epoch it can score feasible masks with the next-step forecast loss, but it does not optimize the consequences of minimum activation, warm-up, or observation age. PD-PPO has lower mean and macro loss in all 24 seeds (Table 7). The greedy reference also switches at 0.0280 per step, approximately 7.6 times the PD-PPO rate. Thus,

Table 10. Matched objective and learned-policy controls. Positive differences indicate lower loss for forecast-reward PD-PPO.

Control	Mean loss: wins; difference [95% CI]	Macro: wins; difference [95% CI]	Macro vs. fixed	Valid trace
AoI reward	10/24; -0.0009 [-0.0074, +0.0052]	13/24; +0.0010 [-0.0039, +0.0058]	24/24	24/24
Diagonal-uncertainty reward	11/24; +0.0001 [-0.0065, +0.0065]	12/24; +0.0008 [-0.0036, +0.0046]	24/24	24/24
Matched Double-DQN	23/24; +0.1408 [+0.1041, +0.1790]	24/24; +0.0697 [+0.0539, +0.0855]	12/24	21/24

Reward controls use the same PPO architecture, observation, training guide, auxiliary context task, and feasible masks; each difference is control loss minus forecast-reward loss. Double-DQN uses the same state, forecast reward, and masks, and its differences are DQN loss minus PD-PPO loss. The last column applies the reward-control execution check or the full DQN action-trace gate. All policies use the common frozen evaluator.

immediate access to a forecast score does not recover the lower-loss, lower-switching sequence learned by the policy.

The warning-score rule and exact-label replay address a different question. The former maps the supplied synthetic warning scores directly to masks; the latter receives held-out regime labels and replays a validation-fitted mapping. Their mean macro differences from PD-PPO are +0.0012 and +0.0018, respectively, and both bootstrap intervals cross zero. The mean-loss comparisons are similarly inconclusive. Direct event information therefore captures much of the available value in this controlled setting, while neither reference establishes a consistent advantage over the learned online policy. Exact-label replay is an information-privileged diagnostic, not an upper bound on feasible sequential control.

6.5. Independent forecaster check

The final trajectories were also scored by an independently fitted ridge forecaster. Static-mask selection was repeated on the validation partition for this evaluator before final scoring. PD-PPO beats the original fixed schedule in 24/24 macro comparisons and the ridge-selected fixed schedule in 23/24 (Table 11). The latter mean margin is +0.1334, with a 95% bootstrap interval of [+0.1112, +0.1544]. This check changes both the forecaster class and the selected fixed reference without retraining the policy.

Table 11. Frozen-trajectory sensitivity under an independently fitted ridge forecaster. Positive macro margins favor PD-PPO.

Fixed-schedule reference	PD-PPO wins	Macro margin [95% CI]
Original validation-selected fixed schedule	24/24	+0.1686 [+0.1362, +0.2053]
Ridge-validation-selected fixed schedule	23/24	+0.1334 [+0.1112, +0.1544]

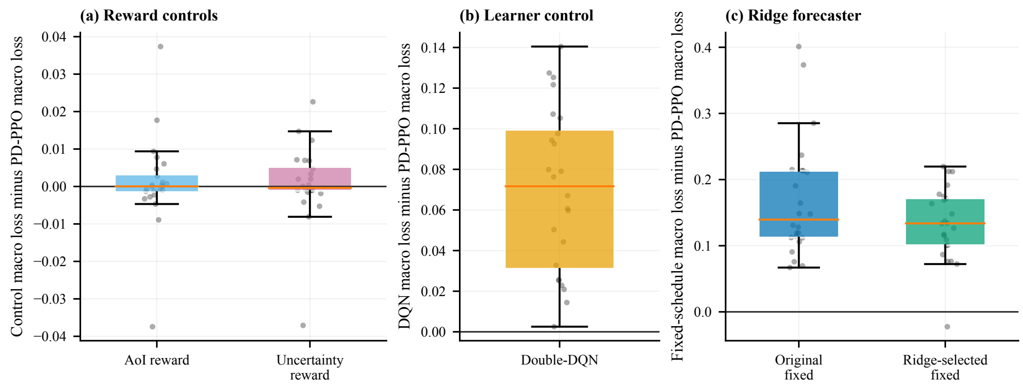

Figure 4. Matched controls and forecaster sensitivity. Panel A compares the forecast-reward policy with AoI- and uncertainty-reward PPO policies. Panel B shows Double-DQN macro loss minus PD-PPO macro loss. Panel C gives the macro margin under the ridge forecaster for the original and ridge-selected fixed schedules. The reward-control intervals include zero, whereas the learner and secondary-forecaster checks retain positive mean margins for PD-PPO.

7. Discussion

7.1. What the evidence establishes

The experiments show that prediction-driven constrained reinforcement learning can allocate a scarce measurement slot more effectively than a fixed design in the reported sensing system. The result is consistent across mean forecast loss and an equal-weight regime score, across all 24 final seeds, and across the unchanged 22-seed post-pilot expansion. It also survives replacement of the evaluation forecaster and comparison with a fixed schedule selected specifically for that forecaster. Continuous replay over all scoreable final-partition epochs retains the fixed-schedule win in every seed, showing that the direction is not created by the primary window-selection rule. The action traces supply the behavioral link: PD-PPO selects the particle, flux, and thermal specialists in the regimes where their measurements are useful, while retaining a broad mixture during calm periods.

This evidence concerns a fixed-backbone, one-specialist instance, but the underlying design problem is broader. Whenever only $r < K$ specialist channels can operate, a scheduler must decide whether to reserve the available capacity for one generally useful measurement or reallocate it as forecast requirements change. The present $r = 1$ setting makes that trade-off directly observable. The weather backbone costs 0.25 normalized units and a useful specialist costs 0.49; one specialist fits within the 0.75 budget, whereas two do not. The gain therefore comes from choosing among mutually exclusive measurements, not from reducing power below the prescribed budget.

7.2. *Why adaptation improves on a strong fixed schedule*

Validation chooses a strong fixed specialist: FC4 is selected in more than half of the seeds, and surface infrared in most of the remainder. Such a schedule avoids switching and can perform well when the evaluation distribution is dominated by one measurement requirement. Its weakness appears when particle, flux, and thermal windows all contribute to the forecasting objective. The specialist preferred in one regime is nearly absent from another, so no single mask can preserve the three regime-specific benefits. This is the empirical case described by Proposition 1.

PD-PPO obtains its margin with sparse switching. Its mean rate of 0.00368 switches per step is compatible with the minimum activation rule and is far below the one-step greedy rate. The policy holds a specialist through a useful interval and changes it when the online state and warning scores indicate a different forecast requirement. Intermediate duty for all four useful specialists, nonzero regime–mask mutual information, and the absence of fixed or cyclic traces distinguish this behavior from both a hidden static solution and mechanical rotation.

7.3. What the matched controls add

The reward controls refine the interpretation of "prediction driven." Forecast loss, AoI, and diagonal uncertainty define different objectives, as shown by Proposition 2; that mathematical distinction does not imply that they must produce different policies in every environment. Here, all three PPO variants receive the same event-weighted training structure, guide, and auxiliary context labels, and the feasible surface contains only four meaningful specialist choices. The paired intervals include zero, so this experiment detects no mean difference among the three reward variants. Forecast loss remains the direct task objective and removes the need to assume that freshness or uncertainty is an adequate surrogate, but the present experiment does not attribute the full performance margin to that scalar reward alone.

The learner control addresses a separate issue. Double-DQN shares the online state, forecast reward, and feasible masks, yet it solves the final problem less consistently than PD-PPO. Its behavior is valid in most seeds and it incurs no warm-up aborts, so the comparison is not explained by gross constraint failure. The result favors the complete masked PPO training package for this small, history-dependent action surface. Because the PPO package also contains a behavior-cloning warm start and an auxiliary classifier, the comparison should not be read as a general theorem that policy-gradient learning dominates value-based learning.

The one-step greedy result provides stronger evidence for sequential value. The greedy method can inspect an immediate future-loss score and therefore has more information per decision than the deployed policy. It still loses in every seed and switches about 7.6 times as often. Minimum activation, warm-up delay, and the persistence of stale observations couple consecutive actions; choosing the best mask for the next step can leave a poor feasible state for the following steps. PD-PPO learns this temporal trade-off without running the forecasting model online.

7.4. Role of event information

The warning-score and exact-label comparisons locate the remaining difficulty. The supplied warning scores are leading signals generated from the synthetic event process, so a validation-fitted threshold rule can react without learning a long credit-assignment chain. Exact-label replay removes warning uncertainty altogether. Neither comparison detects a mean difference from PD-PPO, which shows that explicit regime information explains much of the attainable value in this simulation. It also shows that the learned policy reaches that level while combining noisy warning scores with partial observations, AoI, runtime, and its previous action.

The comparison clarifies what the learned scheduler replaces. Both diagnostics depend on explicit, validation-fitted mappings from warning or regime types to specialists; the exact-label version additionally uses information unavailable during normal operation. PD-PPO learns a feasible sequential mapping from the online state and can retain information from observations between warning transitions. The comparison identifies event-signal quality as an important design variable rather than a hidden source of superiority.

7.5. Online computation and scaling

The frozen forecaster is used for training and offline evaluation. Online execution requires one actor forward pass followed by deterministic masking of actions that violate the power, startup, or minimum activation rule. It neither simulates future trajectories nor evaluates every candidate with a forecasting model.

In the six-channel system, the categorical actor chooses among six predefined masks and has negligible search overhead relative to sensor sampling and estimation.

With several simultaneous specialist slots, explicit enumeration can grow combinatorially. The formulation can retain the same downstream forecast loss and feasibility interface while replacing the categorical head with sequential channel ranking, a factorized subset policy, or a constrained projector. The appropriate parameterization depends on the number of channels and on whether several specialist combinations carry distinct forecast value.

7.6. Limitations

The study uses a controlled generator calibrated to cold-region monitoring, including synthetic event warnings and six logical channels. The 24 seeds cover new event realizations from this generator, not different sites, hardware assemblies, or warning systems. Field use requires measured power and warm-up profiles, warning signals derived from available instruments, and evaluation under real missingness and faults. The exact-label reference remains an offline diagnostic and is not part of the deployed observation interface.

The primary score emphasizes prespecified transport-rich windows so that the three specialist requirements receive comparable weight. Continuous replay over the complete scoreable final partition confirms the same paired direction, but it does not replace evaluation across different sites or event-generation processes.

Schedule comparisons use frozen forecasters. The independent ridge analysis shows that the fixed-schedule result is not confined to one evaluator class, but it re-scores existing trajectories rather than jointly retraining a forecaster for every policy. The reward controls also retain the same training guide and auxiliary labels, which can reduce observable differences among scalar reward proxies. These boundaries motivate hardware replay, multiple channel sets, and joint scheduler–forecaster adaptation as the next validation steps.

8. Conclusion

PD-PPO formulates sensing system scheduling under a power budget as downstream forecast optimization over executable channel subsets. A masked PPO policy combines partial observations, channel ready-sampling age, sensor runtime, prior actions, and early warning scores while enforcing power, startup, and minimum activation rules at the action interface. In the six-channel simulation, it lowers mean forecast loss by 10.1% and the regime-balanced macro score by 7.9% relative to validation-selected fixed schedules, with positive paired results in all 24 seeds. The same 24/24 direction remains when the frozen schedules are replayed over all 5,242 scoreable epochs of each final partition. The learned traces allocate different specialists to particle, flux, and thermal regimes without fixed, cyclic, or high-frequency switching behavior.

Matched controls clarify the source and scope of this result. PD-PPO is more consistent than Double-DQN on the same state, reward, and feasible action surface, and its fixed-schedule advantage persists under an independent ridge forecaster. The matched reward-control intervals include zero in this setting, while direct warning-score and exact-label references show that event information accounts for much of the available scheduling value. Together, the results support masked sequential learning as a practical way to translate forecast requirements into feasible sensor activation decisions when different operating regimes require incompatible measurements.

Appendix A. Supplementary Theoretical Details

Appendix A.1. Proof of Proposition 1

Let $\rho_c > 0$ be the macro weight of regime $c \in C$. A fixed policy chooses one specialist S_0 for every regime. By assumption, no such choice matches the best specialist for all positive-weight regimes. A regime-informed policy π^{lab} observes c_t and changes to the corresponding specialist when the common startup and minimum-activation rules permit. The regime intervals are assumed to be long enough for these changes, and their transition losses are included in the regime losses. Direct regime labels are not available to the deployed learned policy (Table 4).

For a fixed subset S_0, let $\delta_c(S_0)$ be its normalized regime loss minus that of π^{lab}, after transition losses are included, and define the mismatched regime set

$$C_{\mathrm{mis}}(S_0) = \{c \in C \mid S_0 \text{ mismatches the best specialist in } c\}. \tag{A.1}$$

By assumption, $\delta_c(S_0) \geq 0$ in every regime and $\delta_c(S_0) \geq \Delta_c > 0$ for each mismatch. The macro loss of the fixed subset therefore satisfies

$$\mathcal{M}_{\mathrm{macro}}(\pi_{S_0}) - \mathcal{M}_{\mathrm{macro}}(\pi^{\mathrm{lab}}) = \sum_{c \in C} \rho_c \delta_c(S_0) \geq \sum_{c \in C_{\mathrm{mis}}(S_0)} \rho_c \Delta_c > 0. \tag{A.2}$$

The strict inequality follows because $C_{\mathrm{mis}}(S_0)$ is non-empty and every included term is positive. Thus every fixed specialist has higher macro loss under the stated sufficient conditions. The implemented exact-label replay uses a validation-selected regime-to-mask map and the same feasibility interface. It is related to this construction, but it is not assumed to find each regime-optimal feasible policy and is not used as a mathematical upper bound.

Appendix A.2. Proof of Proposition 2

The proof sketch in the main text uses a two-variable construction to show that AoI or covariance proxies can order schedules differently from forecast loss. The construction below uses a linear-Gaussian model to make the ordering explicit. The result is a sufficient-condition argument showing that the ordering can differ, not that it must differ in every system. The experiment uses a nonlinear simulator, but the qualitative mechanism—event-sensitive channels dominating forecast loss during event periods—is present in the simulated data (Table 9). Let the forecast target be

$$y_{t+1} = cz_t + Me_t + \eta_t, \tag{A.3}$$

where z_t is a slowly varying background variable, $e_t \in \{0, 1\}$ is a rare-event indicator, η_t is zero-mean noise, and $M \gg c$. Policy π_1 refreshes the low-power background sensor at every step and never activates the delayed event sensor. Policy π_2 allows the background measurement to become older but activates the event sensor before event onset.

For non-event times, π_1 can have lower AoI and lower covariance trace because the background variable is fresher. During event times, the loss gap in the event term is proportional to M. For sufficiently large M, the expected forecast-loss reduction obtained by observing e_t during events dominates the extra background error induced by the older z_t measurement. Therefore π_1 can have lower AoI or lower linear-Gaussian covariance trace, while π_2 has strictly lower forecast loss. Thus AoI/covariance proxies and the forecast reward can induce different orderings in delayed-activation event environments.

Appendix A.3. Scheduling from state evidence and simple cycles

A fixed cycle chooses specialists from the time index alone. Such a cycle can be optimal only in the special case where regime arrivals are deterministic and phase-aligned with the cycle. If regimes are not locked to the same period and phase, then the cycle has nonzero probability of assigning a mismatched specialist to at least one positive weight regime. Under the same positive mismatch loss assumption used above, that nonzero mismatch probability produces positive expected regret relative to a policy that conditions on informative regime evidence. For this reason, the empirical protocol reports action-trace checks, since the forecast score alone is not enough to distinguish a policy conditioned on state evidence from a fixed schedule or a state independent rotation.

In the reported simulation, blowing-snow events are generated by a semi-Markov process coupled to wind exceedance (Section 5.2), which produces stochastic, non-periodic regime arrivals. The cycle-optimality condition is therefore not satisfied. The action-trace checks in Section 6 confirm that the learned policy is not a simple cycle. Event–mask mutual information is positive in every seed, and the policies do not collapse to the cycle-like patterns excluded by the audit rules in Section 5.

Appendix A.4. Relation to Fixed Subset Allocation

Proposition 2 explains why AoI and covariance objectives can fail, while still allowing fixed subset allocation to be strong. If one fixed specialist covers the variables that dominate mean forecast loss, then a validation-selected fixed policy can match or exceed a learned scheduler on the mean forecast metric. The fixed-backbone, single-specialist study therefore uses the single specialist question in Proposition 1, asking whether a dynamic scheduler can improve particle, flux, and thermal event regimes after normalizing each regime by the difficulty of fixed schedules. This explains why the paper reports the fixed schedule step margin comparison and the macro comparison normalized by fixed-schedule references separately.

Appendix B. Physical Platform Motivation

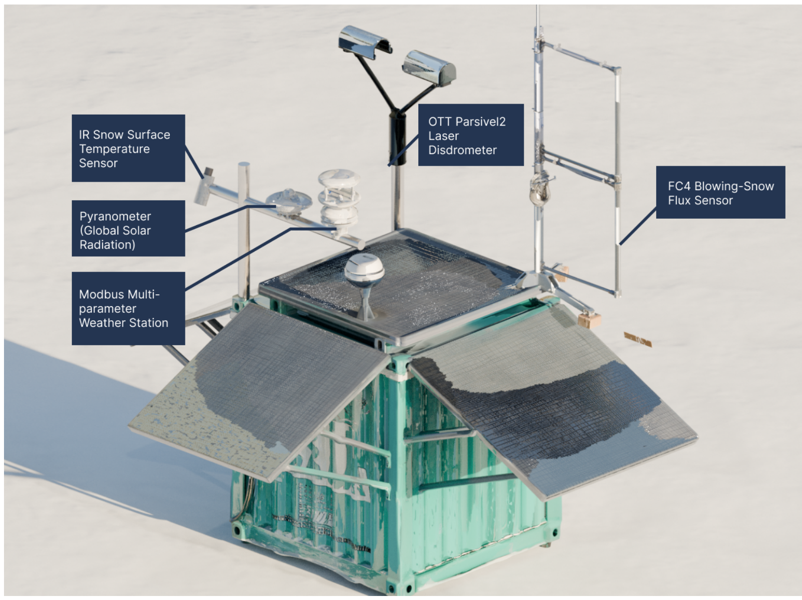

Figure B.5. Author-rendered AWS platform motivating the reported sensing abstraction. The main experiment uses the logical channel table in Table 6 as the experimental definition.

Appendix C. Protocol Hyperparameters

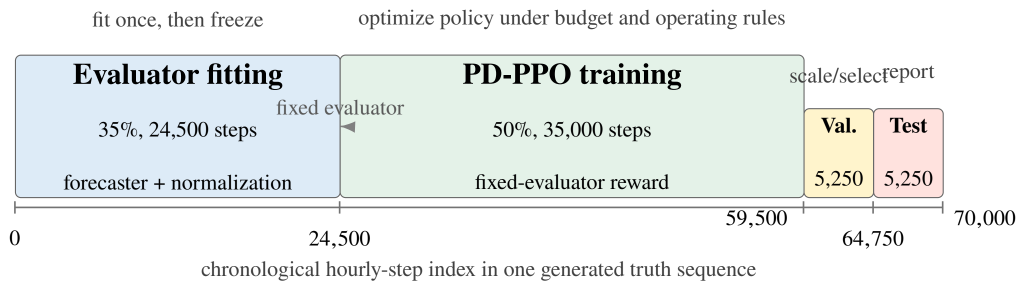

Figure C.6. Chronological split used in the experiments. Forecaster fitting, policy learning, calibration and fixed-schedule selection, and final testing occupy separate intervals. The calibration interval fixes subtype scales and the comparison mask; the final interval supplies the prespecified primary windows and the complete-partition sensitivity replay.

Table C.12. Main protocol and PD-PPO hyperparameters for the reported experiment.

Item	Value
Final seeds	117–140
Truth length	70,000 hourly epochs per seed
Chronological partitions	[0,24500) forecaster fitting; [24500,59500) policy learning; [59500,64750) calibration and fixed-schedule selection; [64750,70000) final test
Forecast lookback / horizon	20 / 8 epochs; uniform horizon weights
Primary evaluation	8 prespecified subtype-balanced, transport-rich windows, 512 steps each; window selection uses no policy losses
Coverage sensitivity	Continuous replay over all 5,242 scoreable final-test epochs; same frozen policies, masks, and normalizers
Candidate masks	6
Required channel	weather backbone
Power / startup limits	0.75 / 0.95
Policy timesteps	200,000
PPO rollout / batch / epochs	1024 / 128 / 8
Policy history indicators	16-step normalized values, slopes, and observation coverage; 10 features
Actor/critic network	two 128-unit observation encoders blended by the aggregate warning; 32-dim candidate-mask embeddings; separate warning-aware critic
Learning rate, discount, GAE	0.0003, 0.99, 0.95
Clip, entropy, value coef.	0.2, 0.0075, 0.5
Imitation and auxiliary weights	$\beta_{\text{BC}} = 2.5$; $\beta_{\text{event}} = 5.0$
Event loss weights	particle/flux/thermal = 1.55/0.95/1.65; calm = 1.0
Training loss scale	subtype-specific median over calibration fixed candidates
Minimum activation time and switch cost	6 epochs; 0.001
Fixed-schedule selection score	event-regime macro loss normalized by fixed schedules
Event-type mix	particle/flux/thermal = 0.40/0.20/0.40
Event coverage target	0.68
Statistical resampling	100,000 percentile-bootstrap samples over paired seed margins; seed 20,260,718
Behavior-cloning pretraining	10,000 steps; 18 epochs; batch size 256

Partitions are fixed before final testing; fixed-schedule selection and loss normalizers use only the calibration partition.

Table C.13. Matched controls and fixed-forecaster fitting protocol.

Item	Value
Reward controls	Same masked PPO and training-only guide; replace forecast loss with normalized target AoI or bounded diagonal uncertainty
Double-DQN control	Dueling 128-unit network; 200,000 steps; replay 100,000; learning starts 5,000; batch 128; 3-step return; target update 1,000; ϵ 1.0 to 0.05 over 20%; learning rate 10^{-4}
Fixed forecaster architecture	residual TCN, 3 levels, 64 channels, kernel size 3, dropout 0.05
Fixed forecaster fitting mixture	78 rollouts / 53,274 rollout epochs: full observation 18; feasible static, round robin, AoI, and random 6 each; six fixed candidate masks 6 each
Normalization	forecaster input/target scales fitted before policy updates; fixed-schedule reference scales fitted on validation windows only
Checkpoint/schedule selection	final policy checkpoint fixed after policy learning; fixed schedule selected by lowest validation macro score
Computation	GPU policy optimization; common CPU forecaster for all final held-out metrics

Appendix D. Simulator and Observation Specification

The meteorological backbone is generated by phase-randomizing contiguous Antarctic AWS anchor series and matching the empirical marginal distribution of each variable. An alternating calm/storm semi-Markov process supplies longer regimes. Its durations are sampled from a lognormal distribution, after which event pulses with bounded duration and gap are placed inside storm intervals. Wind credibility and hysteresis convert the candidate pulses into final event intervals. One of the particle, flux, or thermal subtypes is then sampled for each complete event run.

Within an event run, each subtype latent follows a masked first-order low-pass process

$$z_{t,c} = (1 - \alpha)z_{t-1,c} + \alpha\epsilon_{t,c}, \qquad \epsilon_{t,c} \sim \mathcal{N}(0, 1), \tag{D.1}$$

and is standardized over its active interval. Subtype effects enter the target variables after a six-epoch lag. Table D.14 lists the generator values used in the reported experiment.

Table D.15 records the target scales and task weights used by both the frozen TCN and ridge evaluations. The event subtype selects a predeclared weight vector after each trajectory has been generated; it never changes the online observation supplied to a scheduling policy.

For each subtype c, the supplied warning score is

$$q_{t,c} = \mathrm{clip}_{[0,1]} \left(b_{t,c} + \varepsilon_{t,c}\right), \tag{D.2}$$

where $b_{t,c} = 1$ during the subtype interval, increases linearly during the 16 epochs before onset, and is zero otherwise; $\varepsilon_{t,c}$ is Gaussian noise with standard deviation 0.01. The aggregate warning is $\max_c q_{t,c}$. PD-PPO receives these four warning

Table D.14. Truth-generator parameters used in the reported simulation. One epoch equals one hour.

Component	Value	Role
Meteorological anchor	Panda100, Panda200, Taishan	Phase-randomized Antarctic AWS series with empirical marginal matching.
Sequence length and interval	70,000; 1 h	One independently generated truth sequence per seed.
Event process	alternating semi-Markov storm/calm states	Lognormal state duration with log-scale standard deviation 0.35 and a minimum duration of 24 epochs.
Storm / within-storm event duty	0.65 / 0.78	Derived from the event-coverage target after clipping.
Median storm / calm duration	1,440 / 775 epochs	Parameters of the alternating macro-regime process.
Event pulse duration / gap	Uniform 28–80 / 8–10 epochs	Bursty blowing-snow episodes within storm states.
Wind thresholds and event margin	8 / 12 / 1.4 m s^{-1}	Weak threshold, strong threshold, and event wind-floor margin.
Hysteresis on / off	0.60 / 0.30	Converts candidate pulses and wind credibility into event intervals.
Particle / flux / thermal probability	0.40 / 0.20 / 0.40	One subtype is sampled for each complete event run.
Subtype latent smoothing	0.65	Low-pass coefficient for subtype-specific latent processes.
Target-effect lag	6 epochs	Delay from subtype latent evidence to target effects.
Particle latent scales	0.62 mm; 8.2 m s^{-1}	Diameter and particle-velocity effects.
Flux latent map	scale 0.0014; offset 1.25; clip 2.4	Linear subtype effect before clipping.
Thermal latent scale	6.4 °C	Snow-surface-temperature effect.
Warning-score lead / noise	16 epochs / 0.01	Linear pre-event ramp plus Gaussian noise, clipped to $[0, 1]$.

Table D.15. Target scales and regime-dependent weights used by the common forecast-loss evaluator. Zero-weight variables remain forecaster inputs but do not enter the reported loss.

Forecast target	Scale	Calm/default	Particle	Flux	Thermal
Air temperature	5	0.10	0	0	0.22
Snow-surface temperature	5	0.26	0	0	85
Wind speed	5	0.08	0.08	0.04	0.08
Wind direction sine	1	0	0	0	0
Wind direction cosine	1	0	0	0	0
Solar radiation	100	0	0	0	0
Snow mass flux	10^{-4}	4.5	0.5	7	0.2
Particle diameter	0.2	42	105	5	1
Particle velocity	5	42	105	5	1

scores but not the exact subtype or event labels. This is an explicit simulator assumption representing an upstream warning service rather than a validated field detector.

An active and ready channel observes variable j with its configured probability $p_{i,j,c}$ and adds independent Gaussian noise with standard deviation $\sigma_{i,j,c}$. Table D.16 gives the regime-dependent specialist values. The weather backbone uses standard deviations 0.18 m s^{-1} for wind speed, 1.8 degrees for wind direction, 0.18 °C for air temperature, 0.90 percentage points for relative humidity, and 14 Pa for pressure. The radiometer uses 8 W m^{-2}, while the shielded thermo-hygrometer uses 0.06 °C and 0.30 percentage points.

Table D.16. Availability probabilities and Gaussian noise standard deviations for the three event specialists. Entries within a cell follow the variable order shown in the second column. "Other event" means an active non-matching subtype.

Specialist	Variables	Calm (p, σ)	Other event (p, σ)	Matching subtype (p, σ)
Surface infrared	surface temperature; thermal latent	(0.20, 1.40); (0.10, 1.20)	(0.25, 1.60); (0.18, 1.20)	(0.96, 0.07); (0.98, 0.03)
Laser disdrometer	diameter; particle velocity; particle latent	(0.10, 1.20); (0.10, 3.00); (0.05, 1.50)	(0.18, 0.90); (0.18, 2.80); (0.12, 1.20)	(0.96, 0.06); (0.96, 0.16); (0.98, 0.03)
FC4 flux sensor	mass flux; flux latent	(0.10, 5.0×10^{-6}); (0.05, 1.50)	(0.18, 4.0×10^{-6}); (0.12, 1.20)	(0.96, 2.5×10^{-7}); (0.98, 0.03)

Noise is expressed in the physical units of each variable. The latent evidence variables are simulator constructs used to make regime-dependent specialist value explicit; they are distinct from the exact event-type labels.

Appendix E. Executable Baseline Definitions

Table E.17 turns the comparator taxonomy in the main text into executable procedures. The validation-selected fixed action is

$$m_{\text{fixed}} = \arg\min_{m \in \mathcal{A}} \left(\mathcal{M}_{\text{val}}(m), \mathcal{L}_{\text{val}}(m), \text{index}(m) \right), \tag{E.1}$$

with lexicographic ordering. This mask is frozen before final testing. The warning-score rule uses the same validation procedure separately for calm, particle, flux, and thermal windows. At epoch t, it chooses

$$\widehat{c}_t = \begin{cases} \arg\max_c q_{t,c}, & \max_c q_{t,c} \geq 0.5, \\ \text{calm}, & \text{otherwise}, \end{cases} \tag{E.2}$$

and requests the validation-selected mask for $\widehat{c}_t$.

The one-step forecast-greedy diagnostic snapshots the complete environment, executes each candidate once, restores the snapshot, and minimizes the tuple

$$\left(\mathcal{L}_{\text{step}}(t; m), \frac{1}{N} \| m - a_{t-1} \|_1, \text{index}(m) \right). \tag{E.3}$$

It therefore uses future targets from the final window and is a privileged myopic diagnostic, not an online baseline.

Table E.17. Executable definitions of the comparison schedules. All constrained schedules use the same candidate masks, power projector, channel warm-up model, and six-epoch minimum activation time as PD-PPO unless stated otherwise.

Schedule	Online decision	Selection and tie break	Information status
Validation-selected fixed	Replay one constant candidate mask.	Evaluate all six masks on validation windows; minimize the validation-normalized event macro loss, then mean forecast loss.	Online-feasible fixed design; no final labels.
Fixed priority	Assign linearly decreasing scores by channel index at every epoch.	Deterministic projector order.	Online and non-learned.
Round robin	Give the highest scores to a cyclic group of two channels.	Epoch and channel index define the cycle; projection resolves conflicts.	Online and non-learned.
AoI priority	Score each channel by clipped time since its last ready sampling opportunity.	Project the descending score order.	Online and non-learned.
Random priority	Draw independent Gaussian channel scores at each epoch.	Fixed seed per run; project the sampled ranking.	Online stochastic reference.
Masked Double-DQN	Evaluate six Q-values and greedily select the largest currently feasible action at final test.	Double-DQN target with feasible next-action masking; deterministic lowest-index tie break.	Online learned-policy control; no final labels.
Warning-score rule	Choose the largest particle, flux, or thermal warning score when it is at least 0.5; otherwise choose the calm action.	On validation windows, separately minimize calm or subtype loss for each context; break ties by mean forecast loss.	Uses supplied synthetic warning scores; no final labels.
One-step forecast greedy	Simulate every candidate for one epoch and select the smallest next-step fixed-forecaster loss.	Lower switching fraction, then lower action index.	Uses future final-window targets; privileged diagnostic.
Exact-label diagnostic	Choose the validation-selected specialist for the exact current or look-ahead subtype.	Use the first active subtype in the look-ahead window; break validation ties by mean forecast loss.	Uses simulator labels; privileged diagnostic.
Full observation	Request all six channels at every epoch.	No selection.	Deliberately removes the constraints; upper reference only.

Appendix F. Generator Validation Checks

Table F.18 and Figure F.7 report the generator-family validation checks used to bound the simulation domain. The checks document the simulation construction and are not policy selection criteria.

Table F.18. Generator-family validation checks used to bound the simulation domain.

Validation criterion	Value	Acceptance rule	Passed
Wind speed ACF max deviation, lags 1–12 h	0.048	< 0.05	yes
P(event fraction > 0.75) in 512-step windows	0.065 [0.062, 0.067]	> 0.05	yes
P(event fraction < 0.25) in 512-step windows	0.539	> 0.30	yes
Max KS statistic for AntAWS scalar variables	0.0146	< 0.05	yes
Median blowing-snow event duration	18.0 h	12–20 h	yes
Flux–wind log–log slope	2.966	2.5–3.5	yes
Diameter–wind Spearman correlation	−0.544	< -0.30	yes
Wind speed PSD log-MSE, 0.1–4.0 cpd	0.0419	< 0.10	yes

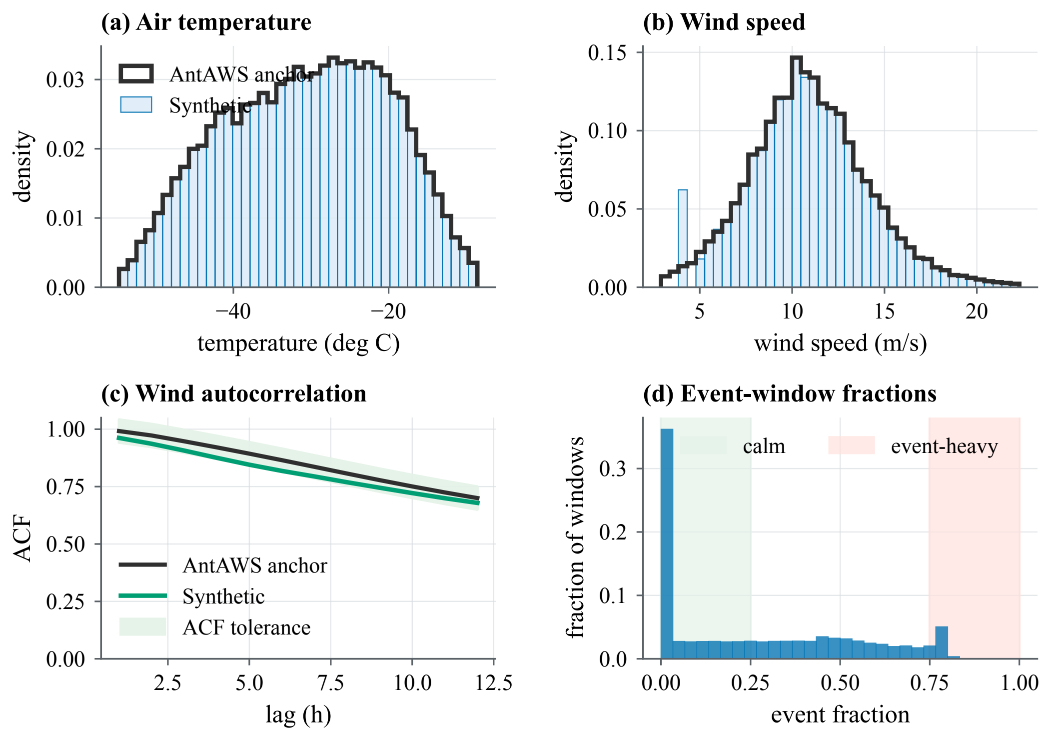

Figure F.7. Statistical checks for the generated truth-sequence family, covering scalar distributions, wind autocorrelation, and event-window heterogeneity.

Appendix G. Fixed-Schedule Selection

Table G.19. Per-seed validation-selected fixed schedule.

Seed	Fixed channels	Score	Seed	Fixed channels	Score
117	Weather backbone + FC4 flux sensor	0.907	129	Weather backbone + FC4 flux sensor	0.892
118	Weather backbone + FC4 flux sensor	0.928	130	Weather backbone + Surface infrared sensor	0.820
119	Weather backbone + Surface infrared sensor	0.804	131	Weather backbone + FC4 flux sensor	0.873
120	Weather backbone + FC4 flux sensor	0.901	132	Weather backbone + FC4 flux sensor	0.915
121	Weather backbone + FC4 flux sensor	0.859	133	Weather backbone + Surface infrared sensor	0.825
122	Weather backbone + Surface infrared sensor	0.844	134	Weather backbone + Surface infrared sensor	0.940
123	Weather backbone + FC4 flux sensor	0.905	135	Weather backbone + Laser disdrometer	0.930
124	Weather backbone + Surface infrared sensor	0.847	136	Weather backbone + FC4 flux sensor	0.892
125	Weather backbone + Surface infrared sensor	0.926	137	Weather backbone + Laser disdrometer	0.880
126	Weather backbone + FC4 flux sensor	0.887	138	Weather backbone + FC4 flux sensor	0.888
127	Weather backbone + FC4 flux sensor	0.904	139	Weather backbone + Surface infrared sensor	0.821
128	Weather backbone + FC4 flux sensor	0.933	140	Weather backbone + Surface infrared sensor	0.905

Scores are validation macro losses; lower is better. The selected schedule is replayed unchanged on final test windows.

CRediT Author Statement

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

A versioned repository containing the code, aggregate tables, seed summaries, generated figure assets, and reproduction scripts will be released with the final manuscript. All reported evidence comes from generated simulation data; undistributed field measurements are outside the evaluation set.

Declaration of Generative AI and AI-Assisted Technologies

During preparation of this manuscript the authors used OpenAI Codex for code inspection, consistency checks, and language-editing assistance. The authors reviewed and edited all generated material and take full responsibility for the content of the manuscript.

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